

DevOps Lab Task - Program 2: Multi-Stage Dockerfile for Node.js

This document demonstrates how to develop a Multi-Stage Dockerfile for building and optimizing a Node.js application container. Multi-stage builds allow the separation of build dependencies from the final runtime image, resulting in smaller, more secure Docker images.

Dockerfile Used in this Program

```
# Stage 1: Build Stage
FROM node:20-alpine AS builder

WORKDIR /app

COPY package.json ./
RUN npm install

COPY . .
RUN npm run build

# Stage 2: Production Stage
FROM node:20-alpine

WORKDIR /app

COPY --from=builder /app/dist ./dist
COPY --from=builder /app/node_modules ./node_modules
COPY --from=builder /app/package.json ./

EXPOSE 3000

CMD ["node", "dist/main.js"]
```

Steps to Build a Multi-Stage Node.js Docker Image

Step 1: List the project files

```
$ ls
```

```
Dockerfile node_modules package.json package-lock.json src dist
```

Step 2: Build the Docker image with a tag

```
$ sudo docker build -t node-layered-image:latest .
```

The build process uses multiple stages to install dependencies, compile the application, and copy only the required files into the final image.

Step 3: Verify the image creation

```
$ sudo docker images
```

You should see the new Node.js image created successfully.

```
(1RV24MC114_VENKATESHAHD@kali)~[/mca/devops/Lab/lab2]
$ ls
dist  Dockerfile  node_modules  package.json  package-lock.json  src
(1RV24MC114_VENKATESHAHD@kali)~[/mca/devops/Lab/lab2]
$ sudo docker build -t node-layered-image .
[+] Building 3.8s (15/15) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 516B
=> [internal] load metadata for docker.io/library/node:18-alpine
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [builder 1/6] FROM docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8ca09d9e
=> [internal] load build context
=> => transferring context: 44.28kB
=> CACHED [builder 2/6] WORKDIR /app
=> CACHED [builder 3/6] COPY package*.json ./
=> CACHED [builder 4/6] RUN npm install
=> [builder 5/6] COPY . .
=> [builder 6/6] RUN npm run build
=> CACHED [stage-1 3/6] COPY --from=builder /app/package.json ./
=> CACHED [stage-1 4/6] COPY --from=builder /app/package-lock.json ./
=> [stage-1 5/6] COPY --from=builder /app/dist ./dist
=> [stage-1 6/6] COPY --from=builder /app/node_modules ./node_modules
=> exporting to image
=> => exporting layers
=> => writing image sha256:dc3038fd0b22ff02aa3cbbdbc626d80dff3420fbd0e6d6e514046536951e219a
=> => naming to docker.io/library/node-layered-image
(1RV24MC114_VENKATESHAHD@kali)~[/mca/devops/Lab/lab2]
$ docker images
permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Head "http://%2Fvar%2Frun%2Fdocker.sock/_ping": dial unix /var/run/docker.sock: connect: permission denied
(1RV24MC114_VENKATESHAHD@kali)~[/mca/devops/Lab/lab2]
$ sudo docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
node-docker-app      latest             707aeff40b17       Less than a second ago   1.14GB
node-layered-image   latest             dc3038fd0b22       9 seconds ago         130MB
flask-image          1.0                8bcfc0b7f1c8       About a minute ago      140MB
flask-image           latest             cdc77d44836d       49 minutes ago         134MB
node                  latest             5cdc8e4df59b       29 hours ago           1.13GB
wordpress            latest             6aaaa8bd9c988      32 hours ago           734MB
nginx                 latest             9d0e6f6199dc       7 days ago             152MB
alpine                latest             706db57f5206       4 weeks ago             8.32MB
ubuntu                latest             97be2d3a3497       5 weeks ago             78.1MB
nimmis/apache-php5   latest             ebf03727818a       17 months ago           44.8MB
registry              2                  26b2eb03618e       2 years ago             25.4MB
mysql/mysql-server    5.6                8c587c89edee       4 years ago             238MB
(1RV24MC114_VENKATESHAHD@kali)~[/mca/devops/Lab/lab2]
$
```